CHAPTER 2

Linear First Order ODEs

1. Use integrating factor method to solve given ODEs.

a) $y' + 3y = t + e^{-2t}$

b) $y' - 2y = t^2 e^{2t}$

c) $y' + y = te^{-t} + 1$

d) $y' + \frac{1}{t}y = 3\cos(2t), \quad t > 0$

e) $y' - 2y = 3e^t$

f) $ty' - y = t^2 e^{-t}, \quad t > 0$

g) $y' + y = 5\sin(2t)$

h) $2y' + y = 3t^2$

2. Solve the given IVPs.

a) $y' - y = 2te^{2t}, \; y(0) = 1$

b) $y' + 2y = te^{-2t}, \; y(1) = 0$

c) $y' + \frac{2}{t}y = \frac{\cos(t)}{t^2}, \; y(\pi) = 0 \;\; t > 0$

d) $ty' + (t+1)y = t, \quad y(\ln 2) = 1, \quad t > 0$

Classification

3. Determine whether the given equation is separable, linear, neither or both.

a) $\frac{dx}{dt} + xt = e^x$

b) $3t = e^t \frac{dy}{dt} + y \ln t$

c) $3r = \frac{dr}{d\theta} - \theta^3$

d) $x^2 \frac{dy}{dx} + \sin x - y = 0$

e) $(t^2 + 1)\frac{dy}{dt} = yt - y$

f) $x\frac{dx}{dt} + t^2 x = \sin(t)$

Separable Equations

4. Determine whether the given equation is separable.

a) $\frac{dy}{dx} - \sin(x + y) = 0$

b) $\dfrac{ds}{dt} = t\ln(s^{2t}) + 8t^2$

c) $(xy^2 + 3y^2)dy - 2xdx = 0$

d) $s^2 + \dfrac{ds}{dt} = \dfrac{s+1}{st}$

e) $\dfrac{dy}{dx} = 4y^2 - 3y + 1$

f) $\dfrac{dy}{dx} = \dfrac{ye^{x+y}}{x^2+2}$

5. Solve the given ODE:

a) $y' = \dfrac{x^2}{y}$

b) $y' + y^2 \sin x = 0$

c) $y' = \cos^2(x)\cos(2y)$

d) $xy' = \sqrt{1-y^2}$

e) $\dfrac{dy}{dx} = \dfrac{x - e^{-x}}{y + e^{y}}$

f) $\dfrac{dy}{dx} = \dfrac{y}{x}$

g) $\dfrac{dy}{dx} = \dfrac{-x}{y}$

h) $\dfrac{dx}{dt} = 3xt^2$

i) $\dfrac{dy}{dx} = \dfrac{x}{y^2\sqrt{1+x}}$

j) $\dfrac{dy}{dx} = \dfrac{\sec^2(y)}{1+x^2}$

k) $\dfrac{dx}{dt} - x^3 = x$

l) $y^{-1}dy + ye^{\cos(x)}\sin(x)dx = 0$

m) $(x + xy^2)dx + e^{x^2}ydy = 0$

n) $x\dfrac{dy}{dx} = \dfrac{1}{y^3}$

o) $\dfrac{dx}{dt} = \dfrac{t}{xe^{t+2x}}$

6. Find a solution to given IVP and determine, if possible, the interval in which the solution is defined.

a) $y' = (1-2x)y^2, \;\; y(0) = -1/6$

b) $y' = \dfrac{1-2x}{y}, \;\; y(1) = -2$

c) $xdx + ye^{-x}dy = 0, \;\; y(0) = 1$

d) $\dfrac{dr}{d\theta} = \dfrac{r^2}{\theta}, \;\; r(1) = 2$

e) $y' = \dfrac{xy^3}{\sqrt{1+x^2}}, \;\; y(0) = 1$

f) $y' = \dfrac{2x}{1+2y}, \;\; y(2) = 0$

g) $y' = \dfrac{3x^2 - e^x}{2y-5}, \;\; y(0) = 1$

h) $\sin(2x)dx + \cos(3y)dy = 0, \;\; y(\pi/2) = \pi/3$

i) $\sqrt{y}dx + (1+x)dy = 0, \;\; y(0) = 1$

j) $\dfrac{1}{\theta}\dfrac{dy}{d\theta} = \dfrac{y\sin(\theta)}{y^2+1}, \;\; y(\pi) = 1$

7. Solve the equation

$$\frac{dy}{dx} = \frac{ay+b}{cy+d}$$

where a, b, c and d are constants.

Existence and Uniqueness

8. Determine (without solving the equation) an interval in which the solution of the given initial value problem is certain to exist. *Hint: Use the theorem* **Existence and Uniqueness, Linear case**.

Pay attention to the form of the equation, if it is not in the form $y' + p(t)y = q(t)$, then first bring it to this form to apply the theorem

a) $(t-3)y' + (\ln t)y = 2t, \;\; y(1) = 2$

b) $y' + (\tan t)y = \sin t, \;\; y(\pi) = 0$

c) $(4-t^2)y' + 2ty = 3t^2, \;\; y(-3) = 1$

d) $y' = \dfrac{2x}{1+2y}, \;\; y(2) = 0$

e) $(\ln t)y' + y = \cot(t), \;\; y(2) = 3$

9. State where in the ty-plane the hypothesis of the theorem **Existence and Uniqueness, Nonlinear case** are satisfied.

a) $y' = \sqrt{1 - t^2 - y^2}$

b) $y' = \dfrac{\ln(ty)}{1 - t^2 + y^2}$

c) $y' + 2ty = (t^2 + y^2)^{3/2}$

d) $y' = \dfrac{1 + t^2}{3y - y^2}$

10. Solve the given initial value problem and determine the interval in which the solution exists depends on the initial value y_0.

a) $y' = \dfrac{-4t}{y}, \;\; y(0) = y_0$

b) $y' = 2ty^2, \;\; y(0) = y_0$

c) $y' + y^3 = 0, \;\; y(0) = y_0$

d) $y' = \dfrac{t^2}{y(1+t^3)}, \;\; y(0) = y_0$

11. By looking at an IVP $dy/dx = f(x, y)$ with $y(x_0) = y_0$, it is not always possible to determine the domain of the solution $y = \phi(x)$ or the interval in which the function $y = \phi(x)$ satisfies the ODE. To see this, consider the following questions.

a) Solve the ODE $dy/dx = xy^3$.

b) Give explicitly the solutions to the IVP with $y(0) = 1$; $y(0) = 1/2$; $y(0) = 2$.

c) Determine the domains of the solutions in part (b).

d) Now solve the ODE with $y(0) = a$, $a > 0$. Show that the domain approaches to whole real line $(-\infty, \infty)$ as a approaches to 0^+ and the domain shrinks to 0 as a approaches to ∞.

12. Let $y = y_1(t)$ be a solution of

$$y' + p(t)y = 0,$$

and let $y = y_2(t)$ be a solution of

$$y' + p(t)y = g(t). \tag{1}$$

Show that $y(t) = cy_1(t) + y_2(t)$ is also a solution to the equation (1) for any constant c.

13. (Method of Separation) The method of separation of variables can give us a solution, but it may not give us all solutions. To illustrate this, consider $dy/dx = y^{1/3}$.

a) Use the method of separation to show that

$$y = \left(\frac{2x}{3} + C\right)^{3/2}$$

is a solution.

b) Show that

$$y = \left(\frac{2x}{3}\right)^{3/2}, \quad x > 0$$

solves the IVP

$$\frac{dy}{dx} = y^{1/3}, \quad y(0) = 0.$$

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c) Show that the constant function $y \equiv 0$ also satisfies the IVP in part (b). Hence this IVP does not have a unique solution.

d) Explain why this does not conflict with the theorem **Existence and Uniqueness, Nonlinear case**.

14. (Method of Separation) To apply method of separation, we might need to divide the given equation by a function of the dependent variable and by dividing we might loose some solutions. To explore this further:

a) Use method of separation to solve

$$\frac{dy}{dx} = (x-3)(y+1)^{2/3}.$$

b) Show that the constant function $y \equiv -1$ also satisfies the above equation.

c) Show that there is no constant c that will make the solution in part (a) yield the solution $y \equiv -1$. Thus, we lost the solution $y \equiv -1$ when we divided by $(y+1)^{2/3}$ to separate variables.

Substitution

15. Sometimes it is possible to solve a nonlinear equation by making a change of variable of the dependent variable that converts it into a linear equation. One important example of such equations is **Bernoulli equation:**

$$y' + p(t)y = q(t)y^n. \tag{2}$$

Solve the following questions about this kind of equations.

a) Solve Bernoulli's equation when $n = 0$ and $n = 1$.

b) Show that if $n \neq 0$, then the substitution $v = y^{1-n}$ reduces the Bernoulli's equation to a linear equation in the function v.

c) Use the method described in part (b) to solve

$$y' = y - y^2.$$

Verify the function you found solves the given equation.

d) Use the method described in part (b) to solve

$$y' + 2y = xy^{-2}.$$

Verify the function you found solves the given equation.

Exact Equations & Method of Integrating Factor

16. Determine whether each of the given equation is exact. If it is exact, find the solution.

a) $(2x + 3) + (2y - 2)y' = 0$

b) $(2x + 4y) + (2x - 2y)y' = 0$

c) $(3x^2 - 2xy + 2) + (6y^2 - x^2 + 3)y' = 0$

d) $\dfrac{dy}{dx} = -\dfrac{ax + by}{bx + cy}$, a, b, c are constants.

e) $\dfrac{dy}{dx} = -\dfrac{ax - by}{bx - cy}$, a, b, c are constants.

f) $(ye^{xy}\cos(2x) - 2e^{xy}\sin(2x) + 2x) + (xe^{xy}\cos(2x) - 3)y' = 0$

g) $(\dfrac{y}{x} + 6x) + (\ln(x) - 2)y' = 0, \quad x > 0$

h) $\dfrac{y}{(x^2 + y^2)^{3/2}} + \dfrac{x}{(x^2 + y^2)^{3/2}}\dfrac{dy}{dx} = 0$

17. Identify each equation as separable, linear, exact, or having an integrating factor that is a function of either x alone or y alone.

a) $(2x + yx^{-1})dx + (xy - 1)dy = 0$

b) $(2y^3 + 2y^2)dx + (3y^2x + 2xy)dy = 0$

c) $(2x + y)dx + (x - 2y)dy = 0$

d) $(y^2 + 2xy)dx - x^2dy = 0$

e) $(x^2\sin x + 4y)dx + xdy = 0$

f) $(2y^2x - y)dx + xdy = 0$

18. Solve the given IVP:

a) $(2x - y) + (2y - x)y' = 0, \quad y(1) = 3$

b) $(9x^2 + y - 1) - (4y - x)y' = 0, \quad y(1) = 0$

19. For each of the following equations find the value of b that makes the equation exact, and then solve it using that value of b.

a) $(xy^2 + bx^2y) + (x^3 + x^2y)y' = 0$

b) $(ye^{2xy} + x) + bxe^{2xy}y' = 0, \quad y(1) = 0$

20. Find an integrating factor of the form $x^n y^m$ and solve the equation.

a) $(2y^2 - 6xy)dx + (3xy - 4x^2)dy = 0$

b) $(12 + 5xy)dx + (6xy^{-1} + 3x^2)dy = 0.$

21. Show that any separable equation

$$M(x) + N(y)y' = 0$$

is also exact.

22. Find an integrating factor and solve the given equation.

a) $(3x^2y + 2xy + y^3) + (x^2 + y^2)y' = 0$

b) $y' = e^{2x} + y - 1$

c) $1 + (x/y - \sin(y))y' = 0$

d) $y + (2xy - e^{-2y})y' = 0$

23. Show that the given equation is not exact but becomes exact when multiplied by the given integrating factor. Then solve the equation.

a) $(x^2 + y^3) + x(1 + y^2)y' = 0, \quad \mu(x, y) = 1/(xy^3)$

b) $(x + 2)\sin(y) + (x\cos(y))y' = 0, \quad \mu(x, y) = xe^x$

c) $(3xy + y^2) + (x^2 + xy)y' = 0$

24. **a)** Prove that $Mdx + Ndy = 0$ has an integrating factor that depends only on the sum $x + y$ if and only if the expression

$$\frac{N_x - M_y}{M - N}$$

depends only on $x + y$.

b) Use part (a) to solve the equation

$$(3 + y + xy)dx + (3 + x + xy)dy = 0.$$

25. Verify that when the linear ODE

$$(p(x)y - q(x))dx + dy = 0$$

is multiplied by $\mu(x) = e^{\int p(x)dx}$, the result is exact.

CHAPTER 3

Homogeneous, Constant Coefficient

1. Find the general solution to given ODE.

a) $y'' + 2y' - 3y = 0$
b) $y'' + 3y' + 2y = 0$
c) $y'' + 5y' = 0$
d) $4y'' - 9y = 0$
e) $y'' - 2y' + 2y = 0$
f) $y'' - 2y' + 6y = 0$
g) $y'' + 2y' + 2y = 0$
h) $y'' + 6y' + 13y = 0$
i) $9y'' + 9y' - 4y = 0$
j) $y'' - 2y' + y = 0$
k) $16y'' + 24y' + 9y = 0$
l) $y'' - 6y' + 9y = 0$
m) $9y'' + 6y' + y = 0$

2. Find the solution to given IVP. Describe the behaviour of the solution as $t \to \infty$.

a) $y'' + y' - 2y = 0; \quad y(0) = 1, \;\; y'(0) = 1$
b) $2y'' + y' - 4y = 0; \quad y(0) = 0, \;\; y'(0) = 1$
c) $y'' + 3y' = 0; \quad y(0) = -2, \;\; y'(0) = 3$
d) $4y'' - y = 0; \quad y(-2) = 1, \;\; y'(-2) = -1$
e) $y'' + 4y = 0; \quad y(0) = 0, \;\; y'(0) = 1$
f) $y'' - 2y' + 5y = 0; \quad y(\pi/2) = 0, \;\; y'(\pi/2) = 2$
g) $y'' + y = 0; \quad y(\pi/3) = 2, \;\; y'(\pi/3) = -4$
h) $y'' + 2y' + 2y = 0; \quad y(\pi/4) = 2, \;\; y'(\pi/4) = -2$
i) $9y'' - 12y' + 4y = 0; \quad y(0) = 2, \;\; y'(0) = -1$
j) $y'' - 6y' + 9y = 0; \quad y(0) = 0, \;\; y'(0) = 2$
k) $y'' + 4y' + 4y = 0; \quad y(-1) = 2, \;\; y'(-1) = 1$

3. Find the ODE whose general solution is $y = c_1 e^{2t} + c_2 e^{-3t}$.

4. Consider the equation $ay'' + by' + cy = 0$, where a, b, c are constants with $a > 0$. Find conditions on a, b and c such that the roots of the characteristic equation are:

(i) real, different, and negative.
(ii) real with opposite sign.
(iii) real, different and positive.

In each case, determine the behaviour of the solution as $t \to \infty$.

Wronskian, Superposition, Linear Independence

5. Find the Wronskian of the given pair of functions.

a) e^{2t}, $e^{-3t/2}$
b) $\cos t$, $\sin t$
c) e^{-2t}, te^{-2t}
d) $e^t \sin t$, $e^t \cos t$
e) $\cos^2 \theta$, $1 + \cos(2\theta)$
f) $e^{\lambda t}\cos(\mu t), e^{\lambda t}\sin(\mu t)$ for μ, λ real numbers.

6. Verify that $y_1(t) = t^2$ and $y_2(t) = t^{-1}$ are two solutions of the differential equation $t^2y'' - 2y = 0$ for $t > 0$. Then show that $y = c_1t^2 + c_2t^{-1}$ is also a solution of this equation for any c_1 and c_2.

7. If $W[f, g](t) = t\cos t - \sin t$, and if $u = f + 3g$, $v = f - g$, find $W[u, v](t)$.

8. Verify that $y_1(t) = 1$ and $y_2(t) = t^{1/2}$ are solutions of the differential equation $yy'' + (y')^2 = 0$ for $t > 0$. Then show that $y = c_1 + c_2t^{1/2}$ is not, in general, a solution of this equation. Explain why this result does not contradict with *Principle of Superposition Theorem.*

9. Show that if $y = \phi(t)$ is a solution of the differential equation $y'' + p(t)y' + q(t)y = g(t)$, where $g(t)$ is not always zero, then $y = c\phi(t)$, where c is any constant other than 1, is not a solution. Explain why this result does not contradict with *Principle of Superposition Theorem.*

10. Find the Wronskian of two solutions of the given differential equations without solving the equation. Hint: Use *Abel's theorem.*

a) $t^2y'' - t(t + 2)y' + (t+)y = 0$

b) $(\cos t)y'' + (\sin t)y' - ty = 0$

c) $(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0$

11. Verify that the functions y_1 and y_2 are solutions of the given differential equation. Do they constitute a fundamental set of solutions?

a) $y'' + 4y = 0; \quad y_1(t) = \cos(2t), \;\; y_2(t) = \sin(2t)$

b) $y'' - 2y' + y = 0; \quad y_1(t) = e^t, \;\; y_2(t) = te^t$

c) $x^2y'' - x(x + 2)y' + (x + 2)y = 0; \; x > 0; \quad y_1(x) = x, \;\; y_2(x) = xe^x$

12. Consider the equation $y'' - y' - 2y = 0$.

a) Show that $y_1(t) = e^{-t}$ and $y_2(t) = e^{2t}$ form a fundamental set of solutions.

b) Let $y_3(t) = -2y_1(t)$, $y_4(t) = y_1(t) + 2y_2(t)$ and $y_5(t) = 2y_1(t) - 2y_3(t)$. Are $y_3(t)$, $y_4(t)$ and $y_5(t)$ also solutions of the given differential equations? Verify your answer.

c) Determine whether each of the following pairs form a fundamental set of solutions: $\{y_1(t), y_3(t)\}$; $\{y_2(t), y_3(t)\}$; $\{y_1(t), y_4(t)\}$; $\{y_4(t), y_5(t)\}$. Explain.

13. Show that if p is differentiable and $p(t) > 0$, then the Wronskian $W(t)$ of two solutions of $(p(t)y')' + q(t)y = 0$ is $c/p(t)$ where c is a constant.

14. Show that $y_1(t) = \cos t$ and $y_2(t) = \sin t$ are a fundamental set of solutions of $y'' + y = 0$; that is, show that they are solutions and their Wronskian is not zero.

15. **(Alternate way to find the second solution)** Consider the equation $y'' + 2ay' + a^2y = 0$.

a) Show that the roots of the characteristic polynomial are $r_1 = r_2 = -a$ so that one solution of the equation is e^{-at}.

b) Use Abel's formula to show that the Wronskian of any two solutions of the given equation is

$$W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t) = c_1e^{-2at},$$

where c_1 is a constant.

c) Let $y_1(t) = e^{at}$ and use the result on part (b) to obtain a differential equation satisfied by the second solution $y_2(t)$. By solving this equation, show that $y_2(t) = te^{at}$.

16. Use the method described in the question (**15.**) or **(Alternate way to find the second solution)** to find a second independent solution of the given equation.

a) $t^2y'' + 3ty' + y = 0,\ t > 0; \quad y_1(t) = t^{-1}$

b) $ty'' - y' + 4t^3y = 0,\ t > 0; \quad y_1(t) = \sin(t^2)$

Reduction of Order

17. Use the method of reduction of order to find a second solution of the given differential equation.

a) $t^2y'' - 4ty' + 6y = 0,\ t > 0; \quad y_1(t) = t^2$

b) $t^2y''2ty' - 2y = 0,\ t > 0; \quad y_1(t) = t$

c) $t^2y'' + 3ty' + y = 0,\ t > 0; \quad y_1(t) = t^{-1}$

d) $ty'' - y' + 4t^3y = 0,\ t > 0; \quad y_1(t) = \sin(t^2)$

Substitution

18. Apply the substitution $x = \ln t$ to solve following equations:

a) $t^2y'' + ty' + y = 0;\ t > 0$

b) $t^2y'' + 4ty' + 2y = 0;\ t > 0$

c) $t^2y'' + 7ty' + 10y = 0;\ t > 0$

d) $4t^2y'' - 8ty' + 9y = 0;\ t > 0$

Existence and Uniqueness, Linear, Homogeneous

19. Determine the largest interval in which the given initial value problem is certain to have a unique twice differentiable solution. Explain why. Do not attempt to find the solution.

a) $y'' + (\cos t)y' + 3(\ln|t|)y = 0, \quad y(2) = 3,\ \ y'(2) = 1$

b) $(x-2)y'' + y' + (x-2)(\tan x)y = 0, \quad y(3) = 1,\ \ y'(3) = 2$

Nonhomogeneous Equations

Method of Undetermined Coefficients

20. Decide whether or not the method of undetermined coefficients can be applied to find a particular solution of the given equation.

a) $y'' + 2y' - y = t^{-1}e^t$

b) $5y'' - 3y' + 2y = -t^3\cos(4t)$

c) $2y'' - 6y' + y = \sin x/e^{4x}$

d) $x'' + 5x' - 3x = 3^t$

e) $y'' + 3y' - y = \sec(\theta)$

f) $y'' + 2y' + y = x^{100}e^{4x}\cos(25x)$

21. Find the general solution of the given ODE by using method of undetermined coefficients.

a) $y'' - 2y' - 3y = 3e^{2t}$

b) $y'' - y' - 2y = -2t + 4t^2$

c) $y'' + y' - 6y = 12e^{3t} + 12e^{-2t}$

d) $y'' - 2y' - 3y = -3te^{-t}$

e) $y'' + 2y' = 3 + 4\sin(2t)$

f) $y'' + 2y' + y = 2e^{-t}$

g) $y'' + y = 3\sin(2t) + t\cos(2t)$

h) $u'' + \omega_0^2 u = \cos(\omega t), \quad \omega^2 \neq \omega_0^2$

i) $u'' + \omega_0^2 u = \cos(\omega_0 t)$

j) $y'' + y' + 4y = 2\sinh t$

k) $y'' - y' + y = (e^t + t)^2$

l) $3y'' + 2y' + 8y = t^2 + 4t - t^2 e^t \sin t$

22. Find the form for a particular solution to

a) $y'' + 2y' - 3y = g(t)$

b) $y'' - 2y' + y = g(t)$

where $g(t)$ is equals to

i) $7\cos(3t)$ **ii)** $2te^t \sin t$ **iii)** $t^2\cos(\pi t)$ **iv)** $5e^{-3t}$ **v)** $3te^t$ **vi)** $t^2 e^t$

23. Find the solution to the given IVP by using method of undetermined coefficients.

a) $y'' + y' - 2y = 2t; \quad y(0) = 0, \;\; y'(0) = 1$

b) $y'' + 4y = t^2 + 3e^t; \quad y(0) = 0, \;\; y'(0) = 2$

c) $y'' - 2y' + y = te^t + 4; \quad y(0) = 1, \;\; y'(0) = 1$

d) $y'' + 4y = 3\sin(2t); \quad y(0) = 2, \;\; y'(0) = -1$

e) $y'' + 2y' + 5y = 4e^{-t}\cos(2t); \quad y(0) = 1, \;\; y'(0) = 0$

24. Determine a suitable form of the particular solution, $Y(t)$, of the given ODE by using method of undetermined coefficients. You are suggested to use a computer algebra system to find the coefficients. (e.g: WolframAlpha)

a) $y'' + 3y' = 2t^4 + t^2 e^{-3t} + \sin(3t)$

b) $y'' - 5y' + 6y = e^t\cos(2t) + e^{2t}(3t + 4)\sin t$

c) $y'' + 2y' + 2y = 3e^{-t} + 2e^{-t}\cos t + 4e^{-t}t^2 \sin t$

d) $y'' + 4y = t^2 \sin(2t) + (6t + 7)\cos(2t)$

e) $y'' + 3y' + 2y = e^t(t^2 + 1)\sin(2t) + 3e^{-t}\cos t + 4e^t$

f) $y'' + 2y' + 5y = 3te^{-t}\cos(2t) - 2te^{-2t}\cos t$

25. Consider the equation

$$y'' - 3t' - 4y = 2e^{-t}.$$

Note that $y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$ are solutions of the corresponding homogeneous equation. Adapting the method of reduction of order, we will seek for a solution of the nonhomogeneous equation of the form $v(t)e^{-t}$. To this end, let $Y(t) = v(t)e^{-t}$.

a) Substitute $Y(t), Y'(t)$ and $Y''(t)$ into given equation to find an ODE v must satisfy.

b) Use the substitution $w(t) = v'(t)$, to solve the ODE you found in part (a) solving for w, and integrating w to find v.

Method of Variation of Parameters

26. Find a particular solution first by undetermined coefficients and then by variation of parameters. Which method was quicker?

a) $y'' - 5y' + 6y = 2e^t$

b) $y'' - y' - 2y = 2e^{-t}$

c) $4y'' - 4y' + y = 16e^{t/2}$

d) $y'' - y = 2t + 4$

e) $2x''(t) - 2x'(t) - 4x(t) = 2e^{2t}$

27. Find the general solution to given differential equation.

a) $y'' + y = \tan t, \quad 0 < t < \pi/2$

b) $y'' + 9y = 9\sec^2(3t), \quad 0 < t < \pi/6$

c) $y'' + 4y' + 4y = t^{-2t}e^{-2t}, \quad t > 0$

d) $4y'' + y = 2\sec(t/2), \quad -\pi < t < \pi$

e) $y'' - 2y' + y = e^t/(1 + t^2)$

f) $y''(\theta) + y(\theta) = \sec^3(\theta)$

g) $v'' + 4v = \sec^4(2t)$

h) $y'' - 5y' + 6y = g(t)$, where g is continuous function.

28. Verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation.

a) $t^2y'' - 2y = 3t^2 - 1, \ t > 0; \quad y_1(t) = t^2, \ y_2(t) = t^{-1}$

b) $t^2y'' - t(t + 2)y' + (t + 2)y = 2t^3, \ t > 0; \quad y_1(t) = t, \ y_2(t) = te^t$

c) $ty'' - (1 + t)y' + y = t^2e^{2t}, \ t > 0; \quad y_1(t) = 1 + t, \ y_2(t) = e^t$

d) $x^2y'' - 3xy' + 4y = x^2 \ln x, \ x > 0; \quad y_1(x) = x^2, \ y_2(t) = x^2 \ln x$

e) $x^2y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 3x^{3/2} \sin x, \ x > 0; \quad y_1(x) = x^{-1/2}, \ y_2(t) = x^{-1/2} \cos x$

29. **a)** By choosing the the lower limit of integration in the Variation of Parameters theorem as the initial point t_0, show that $Y(t)$ becomes

$$Y(t) = \int_{t_0}^{t} \frac{y_1(s)y_2(t) - y_1(t)y_2(s)}{y_1(s)y_2'(s) - y_1'(s)y_2(s)} g(s)ds\,.$$

Show that $Y(t)$ is a solution of the initial value problem

$$L[y] = g(t), \ \ y(t_0) = 0, \ y'(t_0) = 0.$$

b) Show that the solution of the initial value problem

$$L[y] = y'' + p(t)y' + q(t)y = g(t), \ \ y(t_0) = y_0, \ y'(t_0) = y_0'$$

can be written as $y = u + v$, where u and v are solutions of the two initial value problems

$$L[u] = 0, \;\; y(t_0) = y_0, \;\; y'(t_0) = y'_0$$
$$L[v] = g(t), \;\; y(t_0) = 0, \;\; y'(t_0) = 0$$

respectively. In other words, the nonhomogeneitites in the differential equation and in the initial conditions can be dealt with separately. Observe that u is easy to find if a set of fundamental set of solutions of $L[u] = 0$ is known. And, the formula for the function v is given in part (a).

c) Use part (a) and (b) to solve following IVP:

$$y'' + y = g(t), \quad y(0) = y_0, \;\; y'(0) = y'_0,$$

where g is an arbitrary continuous function.

30. The method of reduction of order can also be used for the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t),$$

provided one solution y_1 of the corresponding homogeneous equation is known.

a) Let $y = v(t)y_1(t)$ and show that y satisfies above equation if v is a solution of

$$y_1(t)v'' + (2y'_1(t) + p(t)y_1(t))v' = g(t).$$

Note that this equation is first order in v', so we can solve for v' and integrate the result to find v. Multiplying y_1 with v gives us the general solution. So, this methods simultaneously finds both the second homogeneous equation and a particular solution.

b) Using the method introduced in part (a), solve

$$t^2y'' - 2ty' + 2y = 4t^2, \;\; t > 0; \;\; y_1(t) = t.$$

c) Using the method introduced in part (a), solve

$$t^2y'' + 7ty' + 5y = t, \;\; t > 0; \;\; y_1(t) = t^{-1}.$$

d) Using the method introduced in part (a), solve

$$ty'' - (1 + t)y' + y = t^2e^{2t}, \;\; t > 0; \;\; y_1(t) = 1 + t.$$

CHAPTER 4

Independence

1. Determine whether the given functions are linearly dependent or independent. If they are linearly dependent, find a linear relation among them.

a) $\phi_1(t) = 2t - 3, \;\; \phi_2(t) = t^2 + 1, \;\; \phi_3(t) = 2t^2 - t$

b) $\phi_1(t) = 2t - 3, \;\; \phi_2(t) = 2t^2 + 1, \;\; \phi_3(t) = 3t^2 + t$

c) $\phi_1(t) = 2t - 3, \;\; \phi_2(t) = t^2 + 1, \;\; \phi_3(t) = 2t^2 - t, \;\; \phi_4(t) = t^2 + t + 1$

2. **a)** Show that $W[5, \sin^2 t, \cos(2t)] = 0$ for all t by directly evaluating the Wronskian.

b) Establish the same result without direct evaluation of the Wronskian.

Reduction of Order

3. **a)** Show that if y_1 is a solution of

$$y''' + p_1(t)y'' + p_2(t)y' + p_3(t)y = 0,$$

then the substitution $y = y_1(t)v(t)$ leads to the following second order equation for v':

$$y_1v''' + (3y_1' + p_1y_1)v'' + (3y_1'' + 2p_1y_1' + p_2y_1)v' = 0.$$

b) Use the method of reduction of order as explained in part (a) to solve

$$(2-t)y''' + (2t-3)y'' - ty' + y = 0,\ t < 2;\ \ y_1(t) = e^t.$$

Solving ODEs

4. Verify that the given functions are solutions of the differential equation and determine their Wronskian.

a) $y^{(4)} + y'' = 0;\quad 1,\ t,\ \cos t,\ \sin t$

b) $y''' + 2y'' - y' - 2y = 0;\quad e^t,\ e^{-t},\ e^{-2t}$

c) $xy''' + y'' = 0;\quad 1,\ x,\ x^3$

d) $x^3y''' + x^2y'' - 2xy' + 2y = 0;\quad x,\ x^2,\ 1/x$

5. Find the general solution of the given differential equation.

a) $y''' - y'' - y' + y = 0$

b) $y''' - 3y'' + 3y' - y = 0$

c) $y^{(4)} - 4y''' + 4y'' = 0$

d) $y^{(6)} - 3y^{(4)} + 3y'' - y = 0$

6. Find the solution to the given initial value problem. Determine its behaviour as $t \to \infty$.

a) $y''' + y' = 0;\quad y(0) = 0,\ y'(0) = 1,\ y''(0) = 2$

b) $y^{(4)} - 4y''' + 4y'' = 0;\quad y(1) = -1,\ y'(1) = 2,\ y''(1) = 0, y'''(1) = 0$

c) $2y^{(4)} - y''' - 9y'' + 4y' + 4y = 0;\quad y(0) = -2,\ y'(0) = 0,\ y''(0) = -2,\ y'''(0) = 0$

d) $6y''' + 5y'' + y' = 0;\quad y(0) = -2,\ y'(0) = 2,\ y''(0) = 0$

7. Consider the equation $y^{(4)} - y = 0$.

a) Use Abel's formula to find the Wronskian of a fundamental set of solutions of the given equation.

b) Determine the Wronskian of the solutions $e^t, e^{-t}, \cos t, \sin t$.

c) Determine the Wronskian of the functions $\sin t, \cos t, \sinh t, \cosh t$.

d) Show that the general solution of $y^{(4)} - y = 0$ can be written as

$$y = c_1\cos t + c_2\sin t + c_3\cosh t + c_4\sinh t.$$

e) Determine the solution verifying the initial conditions

$$y(0) = 0,\ y'(0) = 0\ y''(0) = 1,\ y'''(0) = 1.$$

f) Explain why it is convenient to use the functions $\cosh t$ and $\sinh t$ rather than e^t and e^{-t}.

8. Determine the general solution of the given differential equation.

a) $y''' - y'' - y' + y = 2e^{-t} + 3$

b) $y^{(4)} - y = 3t + \cos t$

c) $y''' + y'' + y' + y = e^{-t} + 4t$

d) $y^{(4)} - 4y'' = t^2 + e^t$

e) $y^{(4)} + 2y'' + y = 3 + \cos(2t)$

f) $y^{(6)} + y''' = t$

9. Find the solution of the given initial value problem.

a) $y''' + 4y' = t;\ \ y(0) = y'(0) = 0,\ y''(0) = 1$

b) $y^{(4)} + 2y'' + y = 3t + 4;\ \ y(0) = y'(0) = 0,\ y''(0) = y'''(0) = 1$

c) $y^{(4)} + 2y''' + y'' + 8y' - 12y = 12\sin t - e^{-t}\ \ y(0) = 3,\ y'(0) = 0,\ y''(0) = -1,\ y'''(0) = 2$

10. Determine a suitable form for particular solution $Y(t)$ if the method of undetermined coefficients is to be used. Do not evaluate the coefficients.

a) $y''' - 2y'' + y' = t^3 + 2e^t$

b) $y''' - y' = te^{-t} + 2\cos t$

c) $y^{(4)} - y''' - y'' + y' = t^2 + 4 + t\sin t$

d) $y^{(4)} + 2y''' + 2y'' = 3e^t + 2te^{-t} + e^{-t}\sin t$

CHAPTER 5

Power Series Review

1. Determine radius and interval of convergence for following power series.

a) $\displaystyle\sum_{n=0}^{\infty} \frac{2^{-n}}{n+1}(x-1)^n$

b) $\displaystyle\sum_{n=0}^{\infty} \frac{3^n}{n!}x^n$

c) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n^3}(x-2)^n$

2. Find a power series series expansion of $f'(x)$, given the expansion for $f(x)$.

a) $f(x) = \sin x = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}$